

# Maurizio Ungaro

Research and Software Resume

Staff Scientist, Jefferson Lab | Nuclear physicist and simulation software developer

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## Summary

Staff Scientist at [Jefferson Lab](#) with experience spanning nuclear physics, detector systems, Geant4 simulation, [GEMC](#) development, CLAS12 production workflows, and scientific software infrastructure.

## Core Strengths

- Build and support Geant4-based simulation workflows for detector and radiation-transport studies.
- Develop [GEMC](#) and [CLAS12 Simulations](#) tooling for reproducible geometry, digitization, and production workflows.
- Support CLAS12 detector operations, including LTCC calibration, maintenance, and performance studies.
- Translate complex scientific software into tutorials, examples, technical notes, and user-facing documentation.
- Teach and mentor students in laboratory instruction, data analysis, detector calibration, and scientific computing.

## Selected Experience

### Staff Scientist

2011–present

[Jefferson Lab](#), Hall-B

Geant4 and GEMC simulation support; CLAS12 simulation releases; OSG production workflows; LTCC detector work; nucleon-structure analysis.

### Research Associate

2004–2011

University of Connecticut

CLAS analyses, meson electro-production, CLAS12 software, and detector simulation development.

## Selected Projects

[GEMC](#)

Database-driven Geant4 simulations with a Python-friendly workflow.

[Geant4 at JLab](#)

Tutorials, examples, and support material for Jefferson Lab users.

[CLAS12 Simulations](#)

Tagged simulation releases and production workflows.

[CLAS12 on OSG](#)

Distributed simulation production using HTCondor and Open Science Grid.

[Low Threshold Cherenkov Counter](#)

Forward detector operation, calibration, and performance for pion/kaon discrimination.

[Heavy Photon Search](#)

Dark-sector search collaboration at Jefferson Lab.

[SPRING-8](#)

Prior experimental work in Japan through LEPS/SPRING-8 collaboration.

## Technical Skills

**Simulation:** Geant4, GEMC, CLAS12, ROOT, detector geometry, digitization.

**Programming:** C++, Python, shell, Git, CI, Docker, Meson, CMake, SCons.

**Computing:** HTCondor, OSG, SQLite, CSV/JSON/ROOT workflows, reproducible environments.

**Communication:** documentation, tutorials, talks, mentoring.

## Education

Ph.D. in Nuclear Physics, [Rensselaer Polytechnic Institute](#).

Laurea in Fisica, [Università degli Studi di Genova](#).

## Teaching and Mentoring

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Adjunct Assistant Professor of Physics, Christopher Newport University, 2011. Instructor for PHYS 105 and PHYS 152 laboratories. Teaching assistant at Rensselaer Polytechnic Institute for Physics I and Electromagnetic Theory. Mentor for undergraduate and graduate research in Jefferson Lab data analysis, detector calibration, and CLAS/CLAS12 physics.

## Profiles

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